

Impact Statement

C10 Team Tana

Overview

Our project fine-tunes a small language model (Qwen2.5-1.5B) to generate concise, evidence-grounded answers to patient-facing clinical questions, using retrieval over a fixed corpus of public-health factsheets. The immediate application is narrow short-form guidance across eight topic areas (chronic disease, infectious disease, maternal health, medication safety, nutrition, vaccination, emergency triage, and mental health) but the underlying approach (retrieval-grounded SLMs for domain-specific advice) generalizes to many resource-constrained advisory settings beyond healthcare.

Potential benefits

In communities where frontline health workers frequently operate without immediate physician access, a reliable, source-traceable "second opinion" could meaningfully improve response speed and consistency for common, well-documented conditions. Because every generated answer is linked to a specific retrieved document, users and auditors can verify claims against a named source a transparency property that most general-purpose medical chatbots lack, and one directly relevant to trust and autonomy: a worker can check the model's reasoning rather than accept it on faith.

Potential risks

The most significant risk is misplaced confidence: a fluent, confidently-phrased answer generated from a mismatched or insufficiently specific retrieved document could look authoritative while being subtly wrong, particularly for our emergency_triage and mental_health_basics categories, where errors carry the highest stakes. We observed this failure mode directly during development broadening our retrieval corpus improved topic coverage but degraded answer specificity, since retrieval sometimes selected a topically similar but factually mismatched source. This is a concrete illustration of a risk that could recur, unpredictably, in any deployment with an expanding knowledge base. A second risk concerns equity: our factsheets and training data are English-language and reflect a narrow set of authorial assumptions about how patients phrase questions, which may not represent the linguistic and cultural diversity of communities most in need of this kind of tool.

Uncertainties

We have not tested this system with real users, in a real clinical workflow, or against real (non-synthetic) patient questions, so claims about actual care-access improvement remain speculative rather than demonstrated. We also do not yet know how the model behaves on genuinely out-of-distribution questions, since our current evaluation only covers questions grounded in our fixed 24-document corpus.

Initiatives

We recommend future work on: (1) calibrated confidence signaling, so the model communicates retrieval uncertainty to users rather than presenting all answers with equal confidence; (2) human-in-the-loop evaluation with actual frontline workers before any real deployment; and (3) multilingual, community-sourced factsheet expansion to address the representation gap identified above.